

## Experiments Summary and Result

The first experiment examined how well a Vanilla GAN worked on the MNIST dataset with fully connected generator and discriminator architectures. Training went on for 50 epochs, during which time the generator's loss values were very high initially but then slowly stabilized as the discriminator got used to the changing synthetic samples. The discriminator loss got closer to one after the middle of training, which meant that the adversarial dynamics were balanced. The plotted learning curves showed that the trends between training and validation were the same, which meant that the model was able to generate samples that looked more and more like digits as the adversarial process got better.

The second experiment trained a DCGAN on the CIFAR-10 dataset to see how deeper convolutional architectures affect more colorful and complex image data. The discriminator kept its loss values pretty low over 100 epochs, which showed that it was good at telling the difference between things. The generator's loss, on the other hand, went up over time because CIFAR-10 images are more visually diverse and structurally complex. Even though this kind of training was hard, the adversarial process stayed stable: the generator kept getting better little by little, as shown by the gradual drop in validation loss oscillations. The training curves showed a typical DCGAN learning pattern, with the generator trying to close the performance gap even though the discriminator stayed strong throughout training.

The third experiment in exercise 2 looked at a Universal GAN that had been trained on a custom human-face dataset that had been resized to 128×128 resolution. The model used a generator that could grow with the number of layers and a critic architecture that worked in stages. Across 50 epochs, the discriminator consistently attained low loss values (approximately 0.10–0.30), signifying a high level of confidence in differentiating between real and synthetic samples. In contrast, the generator loss diminished at a slower rate as it adjusted to the intricate structural details found in authentic face images. The training process stayed stable and showed no signs of mode collapse. A qualitative look at the generated samples showed that the model could create coherent facial structures, even though there were still some artifacts that are common in mid-scale GAN training.

In general, the experiments show that the complexity of the model and the difficulty of the dataset have a big effect on how GANs are trained. The Vanilla GAN worked well on data that wasn't too complex, the DCGAN kept making steady but difficult progress on more complex natural images, and the Universal GAN was very flexible with custom real-world data while still keeping training stable. These findings underscore the significance of architectural depth, normalization, and adversarial equilibrium in the scaling of generative models to more intricate tasks.